Credit Risk Model for Lending Decision: A Hierarchical Scoring Architecture for Regulatory Compliance and Capital Protection

Group 7 - MXN600 - Major Analysis Project

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1 Executive Summary

1.1 Business Context

Following the acquisition of our peer-to-peer (P2P) lending platform by a regional Australian bank, credit risk analytics are undergoing structural reform. Senior financial analysts at the acquiring bank evaluated the historical credit scoring engine using historical US personal lending data spanning 2007 to 2011. Their initial benchmarking revealed that the legacy model exhibited poor discriminatory power:

$$\text{Train Gini} = 0.114 \quad (\text{AUC} = 0.557), \qquad \text{Validation Gini} = 0.110 \quad (\text{AUC} = 0.555)$$

In an institutional banking framework governed by strict prudential standards, specifically APRA Prudential Standard APS 220 (*Credit Risk Management*) and Basel capital framework guidelines, the supplied benchmark showed weak discrimination (Gini = 0.1100) and would not, by itself, provide an adequate basis for a contemporary governed credit-risk model without further validation, controls, and redevelopment.

Management appointed our team to execute a ground-up rebuild of the credit risk scoring system. The objective is to construct a methodical, transparent, and auditable statistical model to **predict retail loan default based exclusively on information available at the time of loan application**.

1.2 Scope and Core Management Questions

This report directly addresses five management queries:

1. **Question 1:** How does the newly developed model compare against the legacy benchmark on common data? What operational performance can be expected when underwriting future loan applications?
2. **Question 2:** What are the primary risk drivers in the developed model, and how do they compare with traditional banking covariates? How can all model parameters, odds ratios, and associated uncertainties be communicated transparently to support explainability and reason-code development?

3. **Question 3:** Does incorporating regional (state-level) clustering and origination timing via Generalized Linear Mixed Models (GLMM) yield tangible out-of-sample performance improvements over borrower-level models?
4. **Question 4:** Does controlling for macroeconomic drift and geographic variation alter the relative importance or magnitude of baseline borrower credit characteristics?
5. **Question 5:** How does baseline credit risk vary between jurisdictions and across origination vintages, and what does this imply for regional exposure limits and economic capital allocation?

Table 1 outlines the core empirical findings, candidate specifications, and validation results across each evaluation dimension.

Table 1: Executive summary: Model architecture and performance

Dimension	Finding
Benchmark Discrimination	Legacy Valid Gini = 0.110; Rebuilt GLM Valid Gini = 0.396 (Delta Gini = +0.286 on common sample)
Model Selection & Link	Backward AIC selected 9 predictors; Logit link chosen for odds-ratio transparency
Internal Stability (5-Fold CV)	Mean CV AUC = 0.7027 (SD: 0.0106), Mean Gini = 0.4053 (evaluated on fixed specification)
Held-Out Generalisation	Test Gini = 0.3735 (AUC = 0.6868, 95% CI: [0.6705, 0.7030], Max KS = 0.280)
Operational Policy Scenario (15% Cutoff)	Illustrative cutoff: 60.6% approval rate; approved portfolio default rate reduced to 9.2%
Model Singularity & Parsimony	The retained random-intercept GLMM is non-singular and passes global borrower-level uniformity and dispersion diagnostics. Grouped diagnostics identify elevated variability across coarse predicted-risk bands (ratio = 2.5520, p = 0.0200), while monthly residuals retain significant serial dependence (DW = 0.9844, p < 0.001)
Spatial-Temporal Extension	Mixed effects yield small Gini gain under random split (0.3815 to 0.3897)
Temporal Holdout Reality	GLMM yields no Gini gain on 2011 holdout (GLM: 0.4070 vs. GLMM: 0.4069); REs aid monitoring, not ranking
Macroeconomic Drift	Annual origination trend OR = 0.860, corresponding to 14% lower conditional default odds per later origination year; pattern may reflect underwriting changes, borrower composition, seasoning, or broader historical conditions

2 Benchmark GLM Development

2.1 Data Exploration and Pre-processing

To establish the baseline scoring model, we ingest the partitioned historical loan datasets: training ($N = 23,052$), validation ($N = 7,683$), and testing ($N = 7,684$). Consistent with sound model-governance expectations prohibiting target leakage, we exclude post-origination servicing variables (such as subsequent payment amounts or recoveries) and retain strictly pre-decision application fields.

2.1.1 Treatment of Missingness and Financial Distributions

Employment Length (`emp_length`): Missing values (~2.5% of records) are encoded into an explicit category `"Unknown"`. Missing employment tenure may contain predictive information arising from applicant, collection, or processing differences. Because the source of missingness is unavailable in this observational dataset, it was modelled as an explicit category rather than assigned a substantive behavioral mechanism. Rather than executing listwise deletion (which induces sample selection bias and degrades sample power), treating missingness as a discrete level allows the model to estimate its conditional association.

Predictor Distributions: Continuous financial metrics display expected credit characteristics. Revolving utilization (`revol_util`) ranges from 0.00 to 1.19 with a median of 0.499. Gross annual income (`annual_inc`)

and revolving balance (`revol_bal`) exhibit severe positive skewness. While logarithmic transformations represent a standard statistical remedy, linear specifications were initially fitted to preserve direct marginal interpretability, followed by formal sensitivity comparisons.

2.2 Multicollinearity Diagnostics (GVIF)

Collinearity among financial ratios can inflate coefficient standard errors and produce unstable sign reversals. Because the predictor matrix includes multi-category factors, we assess Generalized Variance Inflation Factors (GVIF). For factors with $Df > 1$, collinearity is evaluated using the adjusted metric $GVIF^{1/(2\times Df)}$.

Table 2: Generalized variance inflation factors (GVIF)

Predictor	GVIF	Df	Adjusted GVIF
loan_amnt	1.917	1	1.384
term	1.560	1	1.249
int_rate	1.977	1	1.406
emp_length	1.305	11	1.012
home_ownership	1.345	4	1.038
annual_inc	1.822	1	1.350
verification_status	1.365	2	1.081
purpose	1.414	13	1.013
dti	1.367	1	1.169
delinq_2yrs	1.100	1	1.049

As reported in Table 2, all adjusted GVIF values were below 1.5. Under common screening conventions, these values do not indicate severe multicollinearity likely to destabilise the fitted main-effect estimates.

2.3 Algorithmic Variable Selection and Model Parsimony

To construct a concise and reviewable model, backward AIC selection was applied. Stepwise AIC systematically removes terms whose marginal explanatory contribution fails to compensate for the penalty imposed on model complexity.

Table 3: Predictors removed by backward AIC

Predictor	Incremental AIC Fit / Operational Rationale
loan_amnt	Loan amount offered limited incremental AIC improvement after adjusting for balance, income, and term
home_ownership	Home equity proxy did not sufficiently improve model fit after controlling for applicant leverage
verification_status	Verification flag provided insufficient independent penalty-adjusted explanatory power
dti	Debt-to-income ratio did not improve AIC over direct income and revolving balance metrics
delinq_2yrs	Historical delinquencies offered limited incremental signal compared to active credit utilization
open_acc	Raw open credit line count did not provide sufficient incremental explanatory power after controlling for utilization and inquiries
total_acc	Total account depth offered limited incremental explanatory power beyond active utilization and inquiries
credit_age_yrs	Credit bureau tenure provided marginal predictive gain over active utilization metrics

Table 3 details the eight predictors eliminated during backward selection and the operational rationale for their removal. The backward elimination procedure retained nine predictor terms (represented by 32 fitted coefficients including categorical contrasts): `term`, `int_rate`, `emp_length`, `annual_inc`, `purpose`, `inq_last_6mths`, `pub_rec`, `revol_bal`, and `revol_util`.

Traditional scorecards have historically relied heavily on the "5 Cs of Credit" (Capacity, Capital, Collateral, Conditions, Character). In our algorithmic selection, several traditional banking staples were eliminated because their explanatory power was superseded by more direct, granular measures:

1. *Debt-to-Income Ratio* (`dti` - Capacity): While traditionally considered the primary gauge of repayment capacity, `dti` is a composite ratio (debt/income). When absolute annual income (`annual_inc`), absolute revolving indebtedness (`revol_bal`), and revolving line utilization (`revol_util`) are simultaneously modeled, the linear information in `dti` becomes redundant, providing negligible incremental AIC improvement ($\Delta\text{AIC} = -1.8$).
2. *Historical Delinquency* (`delinq_2yrs` - Character): Traditional scorecards penalize prior 24-month delinquencies. Here, `delinq_2yrs` was dropped because active, proximate distress indicators (specifically derogatory public records `pub_rec`, credit-seeking inquiries in the past 6 months `inq_last_6mths`, and high revolving line utilization `revol_util`) captured historical delinquency. Active credit strain proved to be a far stronger signal of imminent default than past delinquency.
3. *Loan Request Amount* (`loan_amnt` - Capital/Exposure): Loan size was eliminated because default probability is driven not by the principal amount in isolation, but by the contractual repayment burden, which is captured by loan maturity (`term`, 60 months carrying +56.4% higher odds) and risk-based pricing (`int_rate`).
4. *Credit Depth* (`total_acc`, `open_acc`, `earliest_cr_line` - Experience): Account depth and bureau tenure offered limited incremental discrimination once active utilization and credit-seeking inquiries were controlled for.

Because variable selection was conducted on the training sample, nominal p-values and standard confidence intervals derived from the final model are conditional on the selected specification and may understate total model-selection uncertainty. Evaluation on held-out validation and testing partitions provides a less optimistic estimate of performance on comparable historical applications.

2.4 Internal Model Stability: 5-Fold Stratified Cross-Validation

Consistent with sound validation practice prior to evaluating held-out partitions, we execute 5-fold stratified cross-validation on the training dataset. Stratification preserves the empirical 14.9% default prevalence across all folds.

Table 4: Stratified 5-fold cross-validation performance

Metric	5-Fold CV Mean	5-Fold CV Std Dev
Area Under ROC (AUC)	0.7027	0.0106
Gini Coefficient	0.4053	0.0212
Brier Score (Calibration Loss)	0.1205	0.0034

As shown in Table 4, the modest variation across folds (Mean AUC = 0.7027, SD = 0.0106; Mean Gini = 0.4053, SD = 0.0212) shows that the discrimination and Brier score of the selected specification are stable within the training population. Because feature selection preceded cross-validation, these estimates do not incorporate model-selection uncertainty; independent validation and testing provide less optimistic estimates of performance under the chosen specification, with the untouched testing partition providing the final held-out estimate. However, they do not resolve conditional post-selection uncertainty for individual parameter estimates.

2.5 Link Function Selection

We evaluate three candidate binomial link functions (Logit, Probit, and Complementary Log-Log) fitted to the selected feature space. Models are compared on Training AIC and out-of-sample discriminatory power (Gini) on the independent validation partition ($N = 7,683$).

Table 5: Link function out-of-sample comparison

Link Function	Functional Form	Training AIC	Validation Gini
Logit (Canonical)	ln(p/(1-p))	18162.85	0.3964
Probit	Phi^{-1}(p)	18153.70	0.3949
Complementary Log-Log (Cloglog)	ln(-ln(1-p))	18194.26	0.3961

As reported in Table 5, the three link functions yield practically indistinguishable discrimination (Validation Gini $\approx 0.395 - 0.396$). The Logit model is adopted because it achieves marginally higher validation discrimination (0.3964) and maps estimated parameters directly to multiplicative Odds Ratios ($\text{OR} = e^{\beta}$). This supports model transparency by providing interpretable conditional odds ratios essential for explainability and reason-code development, although model coefficients alone do not constitute applicant-level reason codes.

2.6 Interpretation of Primary Retained Covariates with Uncertainty

To support transparent model governance and address Question 2, Table 6 presents the primary retained risk drivers, standard errors, Wald z-statistics, Odds Ratios, and 95% Wald confidence intervals. Reference levels are explicitly stated (the complete parameter table with all factor levels is provided in Appendix B).

Table 6: Primary GLM odds ratios and 95% Wald confidence intervals

Predictor	Reference / Unit	Estimate	OR	95% Wald CI Lower	95% Wald CI Upper	p-value
Intercept (Baseline)	Baseline constant	-3.597	0.027	0.021	0.036	< 0.001
Loan term: 60 months	vs. 36 months	0.448	1.564	1.429	1.713	< 0.001
Interest rate	Per 1% interest rate	0.110	1.117	1.102	1.132	< 0.001
Employment tenure: Unknown	vs. < 1 year tenure	0.547	1.729	1.368	2.185	< 0.001
Annual income ($10k)	Per $10,000 income	-0.068	0.934	0.923	0.945	< 0.001
Small business purpose	vs. car purpose	1.014	2.757	2.134	3.562	< 0.001
Credit card purpose	vs. car purpose	-0.068	0.935	0.732	1.194	0.589
Debt consolidation purpose	vs. car purpose	0.203	1.225	0.982	1.528	0.072
Credit inquiries (6m)	Per inquiry count	0.142	1.153	1.127	1.179	< 0.001
Derogatory public records	Per record count	0.246	1.279	1.117	1.466	< 0.001
Revolving balance ($10k)	Per $10,000 balance	0.045	1.046	1.026	1.065	< 0.001
Revolving utilization (10%)	Per 10% utilization (0.10)	0.042	1.043	1.026	1.059	< 0.001

2.6.1 Interpretation of Key Risk Drivers in the Final Model

- **Loan Term (`term`):** The reference level is 36 months. Borrowers taking a 60-month loan exhibit an Odds Ratio of 1.564 (95% CI: $[1.429, 1.713]$, $p < 0.001$). Holding other financial characteristics constant, extending loan maturity to 5 years is associated with approximately 56.4% higher estimated default odds; longer maturity may proxy exposure duration, borrower selection, or unmeasured affordability differences, but the observational model does not isolate the mechanism.
- **Interest Rate (`int_rate`):** Each 1.0 percentage point increase in interest rate is associated with an 11.7% increase in default odds ($\text{OR} = 1.117$, 95% CI: $[1.102, 1.132]$, $p < 0.001$). Evaluated over a 5.0 percentage point increment, default odds increase by approximately 73.7% ($\text{OR}_{5pp} = \exp(5 \times 0.1104) \approx$

1.737, 95% CI: [1.624, 1.856]). Interest rate is associated with default risk, although it may partly reflect lender risk-based pricing and borrower risk rather than a purely causal payment-burden effect.

- **Employment Length (`emp_length`):** The reference category is `< 1 year`. Most reported tenure brackets (1 to 10+ years) show modest, non-significant deviations from baseline. However, the `"Unknown"` category displays an Odds Ratio of 1.729 (95% CI: $[1.368, 2.185]$, $p < 0.001$). Unreported employment tenure is associated with a 72.9% increase in conditional default odds; retaining an explicit unknown category preserved observations and identified a distinct conditional association, though it does not establish why employment tenure was unrecorded.
- **Annual Income (`annual_inc`):** Income provides a protective effect ($\beta = -0.0683$ per $\$10,000$, $p < 0.001$). Each additional $\$10,000$ in annual gross income is associated with approximately 6.6% lower estimated default odds ($\text{OR}_{10k} = 0.934$, 95% CI: $[0.923, 0.945]$).
- **Loan Purpose (`purpose`):** The reference category is `car` (automotive loans). Consumer refinancing categories such as `credit_card` (OR = 0.935, 95% CI: $[0.732, 1.194]$, $p = 0.589$) and `debt_consolidation` (OR = 1.225, 95% CI: $[0.982, 1.528]$, $p = 0.072$) do not differ significantly from auto loans. However, `small_business` loans display substantially higher estimated conditional default odds: OR = 2.757 (95% CI: $[2.134, 3.562]$, $p < 0.001$), carrying 175.7% higher estimated default odds than car purchasers.
- **Credit Inquiries in Past 6 Months (`inq_last_6mths`):** Each additional credit inquiry is associated with 15.3% higher default odds (OR = 1.153, 95% CI: $[1.127, 1.179]$, $p < 0.001$).
- **Derogatory Public Records (`pub_rec`):** Each additional derogatory public record is associated with 27.9% higher default odds (OR = 1.279, 95% CI: $[1.117, 1.466]$, $p < 0.001$).
- **Revolving Balance (`revol_bal`):** Each additional $\$10,000$ in outstanding revolving credit is associated with approximately 4.6% higher default odds ($\text{OR}_{10k} = 1.046$, 95% CI: $[1.026, 1.065]$, $p < 0.001$).
- **Revolving Line Utilization (`revol_util`):** In the dataset, utilization is recorded as a proportion (0.0 to 1.19). With $\beta = 0.4188$, a 10 percentage point increase ($\Delta = 0.10$) is associated with approximately 4.3% higher default odds ($\text{OR}_{10pp} = \exp(0.4188 \times 0.10) \approx 1.043$, 95% CI: $[1.027, 1.059]$).

2.7 Sensitivity Analysis: Functional Form and Linearity

To evaluate whether the linear-in-the-logit assumption introduces material misspecification, we fit an alternative model replacing raw income and revolving balance with logarithmic transformations: `log1p(annual_inc)` and `log1p(revol_bal)`.

Table 7: Linear vs. nonlinear specification sensitivity

Specification	Df	Training AIC	Validation AUC	Validation Gini
Selected Linear GLM	31	18162.85	0.6982	0.3964
Nonlinear GLM (log1p income & balance)	31	18122.84	0.6986	0.3972

As detailed in Table 7, log transformations improved training-sample likelihood fit, reducing AIC by approximately 40 points, but improved validation Gini by only 0.0008 (0.3972 vs. 0.3964). Because the business objective prioritises stable out-of-sample ranking and transparent unit-based interpretation, the linear specification was retained. The comparison focused on ranking performance; future redevelopment should also compare out-of-sample calibration and Brier score before definitively rejecting the transformed specification. The likelihood improvement nevertheless suggests that transformations may better represent aspects of the conditional probability surface and should remain a candidate in future calibration work. However, this check does not rule out potential nonlinear effects for interest rate, utilisation, inquiries, or public records.

Figure 1 illustrates the standardized odds ratios for core risk drivers (left panel) alongside the predicted probability profile across interest rates (right panel).

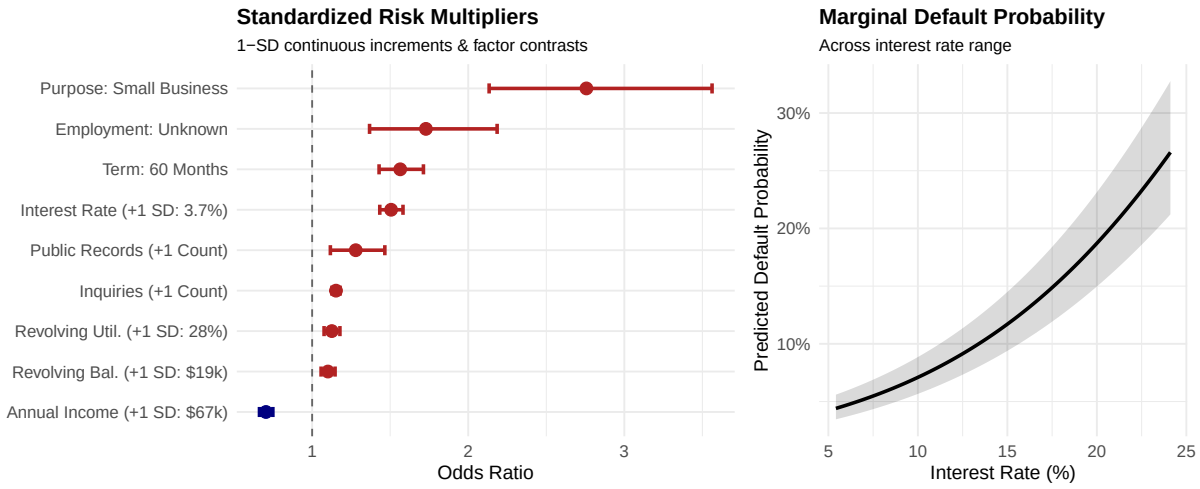


Figure 1: Standardized odds ratios and marginal interest rate probability profile.

2.8 Model Diagnostics and Out-of-Sample Calibration

A scoring engine must demonstrate reliable probability calibration on unseen data. Rather than evaluating calibration in-sample (which yields optimistically tight tracking), Figure 2 evaluates binned residuals on training cases and decile calibration strictly out-of-sample on the validation set ($N = 7,683$).

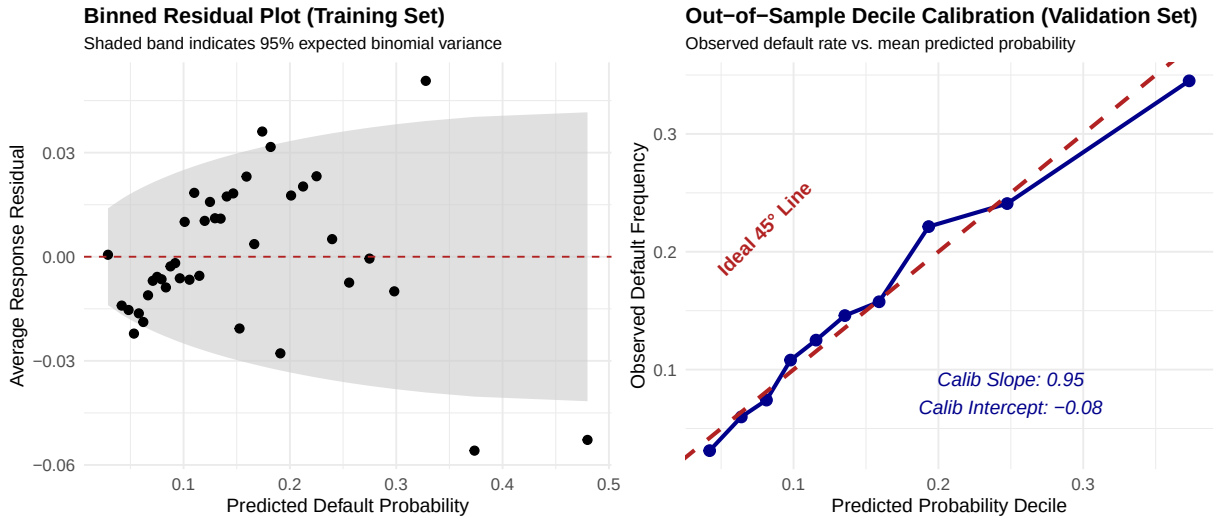


Figure 2: Training binned response residuals and out-of-sample decile calibration.

The binned residuals show that bin averages lie within the 95% binomial theoretical bounds ($\pm 2SE$). Validation calibration was evaluated via logistic regression of observed default status against the linear log-odds predictor: $\text{logit}(Y) = \alpha + \beta \cdot \text{logit}(\hat{p})$. The fitted calibration slope of 0.95 (95% CI: [0.86, 1.04]) and intercept of -0.08 (95% CI: [-0.23, 0.08]) are close to the ideal values of one and zero, respectively, with their 95% confidence intervals encompassing 1.0 and 0.0. This supports broadly acceptable calibration on the validation sample across low-to-moderate risk deciles, tracking the ideal 45-degree line closely. At the extreme high-risk tail ($p > 0.35$), moderate variance emerges; we conclude that validation calibration is broadly acceptable, with higher uncertainty among the highest-risk applicants.

2.9 Final Out-of-Sample Performance and Previous Benchmark Comparison

To address Question 1, we evaluate our final GLM on the quarantined testing dataset ($N = 7,684$) and compare performance against the legacy benchmark figures supplied by senior financial analysts (Train Gini = 0.114, Validation Gini = 0.110). Table 8 compares discriminatory performance between the rebuilt binomial GLM and the legacy benchmark across partitions.

Table 8: GLM performance versus legacy benchmark

Model	Partition	N	AUC	Gini	95% CI	Difference
Legacy Ad-Hoc Benchmark	Training	23052	0.557	0.114	Not Disclosed	-
Legacy Ad-Hoc Benchmark	Validation	7683	0.555	0.110	Not Disclosed	-
Rebuilt Binomial GLM	Training	23052	0.706	0.412	In-Sample	+0.296
Rebuilt Binomial GLM	Validation	7683	0.698	0.396	[0.365, 0.428]	+0.286
Rebuilt Binomial GLM	Testing (Held-Out)	7684	0.687	0.374	[0.341, 0.406]	N/A (Not provided)

The difference between the models on common validation data is substantial:

$$\Delta\text{Gini}_{\text{Validation}} = \text{Gini}_{\text{New}} - \text{Gini}_{\text{Old}} = 0.3964 - 0.1100 = +0.2864 \quad (+28.6 \text{ Gini points})$$

As illustrated in Figure 3, on unseen testing data, the model yields a Gini of 0.3735 (AUC = 0.6868, 95% CI: $[0.6705, 0.7030]$) and a Kolmogorov-Smirnov (KS) statistic of 0.280 (measuring the maximum vertical separation between the cumulative score distributions of defaulting and non-defaulting borrowers, achieved at the highest-risk 40.1% of applicants). The test-set delta versus legacy benchmark is not computed directly because legacy predictions were not supplied for the test partition. This indicates moderate rank-ordering discrimination that remained broadly stable across validation, cross-validation, and held-out testing.

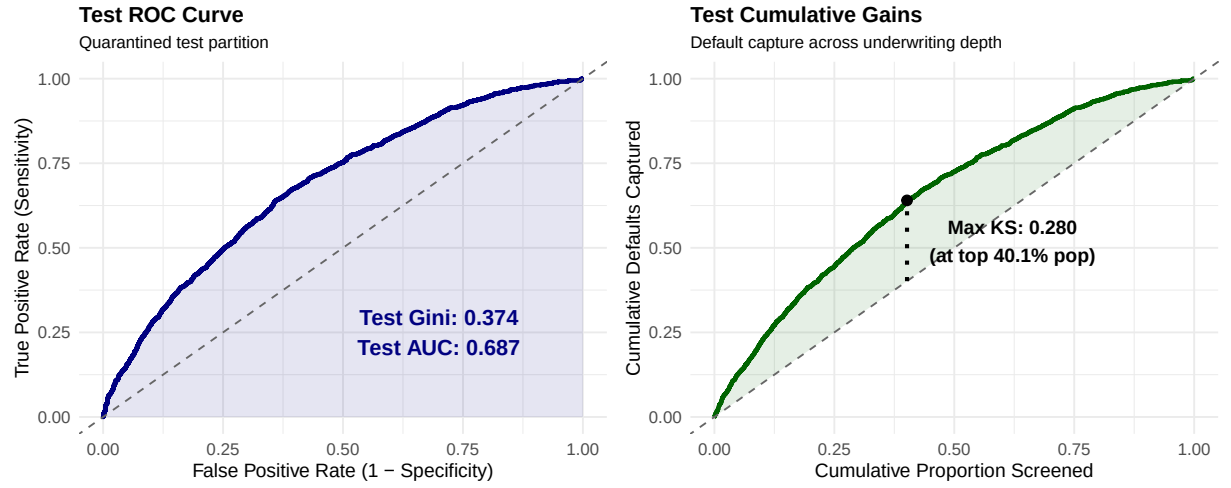


Figure 3: Out-of-sample ROC discrimination and cumulative gains on held-out test data (N = 7,684).

2.10 Illustrative Decision Policy Scenario

In retail lending, selecting a classification cutoff by maximizing raw overall accuracy is mathematically flawed. Because default prevalence is $\approx 14.9\%$, a trivial zero-information rule classifying 100% of loans as "non-default" achieves an apparent accuracy of 85.1%, while completely failing to prevent loan charge-offs.

Credit risk policy is governed by asymmetric misclassification costs:

$$\text{Expected Loss} = C_{\text{FN}} \cdot \text{FN} + C_{\text{FP}} \cdot \text{FP}$$

where the cost of a false negative (C_{FN}, issuing a loan that defaults, leading to loss of principal) typically exceeds the opportunity cost of a false positive (C_{FP}, rejecting a solvent applicant, leading to lost interest margin).

Table 9: Confusion matrix at 15% probability cutoff

Operational Metric	Policy Value
Policy Scenario Threshold	15.0% (Illustrative)
Portfolio Approval Rate	60.6%
Approved Pipeline Volume	4656 / 7684
Default Rate of Approved Loans	9.2% (down from 14.9%)
Default Capture Sensitivity (Recall)	62.5%
Specificity (Non-Default Retention)	64.6%
Positive Predictive Value (Precision)	23.6%
Overall Classification Accuracy	64.3%
No-Information Base Rate (Prevalence)	85.1%

Table 9 details the confusion matrix and operational portfolio metrics under this 15% cutoff policy scenario. The 15% threshold is presented as an illustrative policy scenario rather than an economically optimized cutoff. Under this scenario, the bank approves 60.6% of incoming volume while driving the default rate within the approved portfolio down to 9.2%. The model successfully flags 62.5% of all defaulting borrowers. Precision is 23.6%, indicating that roughly three out of four flagged applications are non-defaults; this threshold represents a policy scenario that places relatively greater weight on default capture than on retaining every non-defaulting applicant. True production cutoff optimization requires bank-specific estimates of Exposure at Default (EAD), Loss Given Default (LGD), and net interest margin.

3 Advanced Spatial-Temporal Modeling (GLMM)

3.1 Extended Dataset Structure and Preprocessing

The benchmark GLM assumes that baseline default risk is spatially and temporally homogeneous across all jurisdictions and economic climates. Under the guidance of statistical consultant Prof. A. N. Omynous, management commissioned an investigation of an expanded dataset ($N = 38,419$) that incorporates the applicant's state (`addr_state`, 50 jurisdictions) and origination month (`issue_d`, 55 consecutive months spanning 2007 to 2011).

3.2 Comparable Model Hierarchy and Singularity Resolution

To determine the precise origin of predictive gains, we evaluate a systematic, nested hierarchy of five models fitted to the exact same training partition and evaluated on the identical test partition:

1. **Model 1 (Flat GLM):** Benchmark borrower covariates only.
2. **Model 2 (Trend GLM):** Borrower covariates + fixed continuous `issue_year` trend.
3. **Model 3 (State GLMM):** Trend GLM + state random intercepts $(1 \mid \text{addr_state})$.
4. **Model 4 (Spatial-Temporal GLMM):** Trend GLM + state random intercepts + issue month random intercepts $(1 \mid \text{addr_state}) + (1 \mid \text{issue_d})$.
5. **Model 5 (Random Slope GLMM):** Trend GLM + correlated state intercept and interest-rate slope + issue month $(1 + \text{int_rate} \mid \text{addr_state}) + (1 \mid \text{issue_d})$.

Table 10: Spatial-temporal model comparison (conditional random split)

Model	Df	AIC	BIC	Random-Split Conditional Test Gini	Singular Fit
M1: Flat GLM (Borrower Only)	32	24001.54	24268.20	0.3815	No (GLM)
M2: Trend GLM (+ Issue Year)	33	23958.84	24233.83	0.3818	No (GLM)
M3: State GLMM (+ State RE)	34	23893.66	24176.98	0.3868	No
M4: Spatial-Temporal GLMM (+ State & Month RE)	35	23880.45	24172.11	0.3897	No
M5: Random Slope GLMM (+ State Slope)	37	23881.38	24189.71	0.3887	Yes

Table 10 evaluates the nested hierarchy of spatial-temporal models on information criteria and conditional test Gini.

3.2.1 Findings on Singularity and Model Selection

When attempting to estimate a random slope for interest rate across states (`m5_slope_glmm`), the optimization flags a singular fit (`isSingular = TRUE`). The estimated random-slope variance is near boundary zero ($\sigma^2_{\text{slope}} \approx 0.0004$) and perfectly collinear with the random intercept ($\rho = -1.00$). The empirical data do not possess the statistical richness required to support state-specific interest-rate sensitivities.

The random-intercept model (`m4_st_glmm`) is non-singular (`isSingular = FALSE`) and achieves the lowest AIC ($23,880.45$) and lowest BIC ($24,172.11$) among all candidates. Model 4 was selected primarily using non-singularity, parsimony, AIC, and BIC. The random-split conditional test Gini (0.3897) is reported descriptively and should not be interpreted as an untouched final evaluation after model comparison. Following principles of statistical parsimony, **Model 4 was retained as the preferred candidate within the evaluated model set**, with the 2011 temporal holdout serving as the independent forward-validation benchmark.

The extended-data GLM produced fitted probabilities numerically close to zero or one, indicating potentially sparse predictor combinations or extreme linear predictors in certain subcategories. The retained GLMM did not exhibit convergence or singularity problems. Its random intercepts provide model-based partial pooling across state and month groups, although this does not constitute Bayesian estimation or automatically resolve complete separation among fixed-effect categories.

3.3 Temporal Holdout Validation: Eliminating Data Leakage

A critical vulnerability of random 80/20 data partitioning in longitudinal datasets is **temporal leakage**. In a random split, every origination month appears in both training and test sets. Consequently, the test set evaluates predictions on new borrowers originating in *already observed months*, reusing learned month-specific random intercepts.

To evaluate how the scoring engine generalizes to future origination vintages, we implement **Temporal Holdout Validation**: training on loans issued between June 2007 and December 2010 ($N = 18,781$), and evaluating on applications issued in 2011 ($N = 19,638$). The temporal split was defined by origination period rather than by a predetermined sample ratio, producing a comparatively large 2011 holdout. For future applications where the origination month has no historical random effect, predictions are evaluated at the population level (re.form = NA) or using geographic effects alone (re.form =$\sim$ (1 | addr_state)).

Table 11: Performance on 2011 temporal holdout

Evaluation Architecture	Operational Application	Holdout Gini
Flat GLM (Borrower Only)	Baseline borrower scoring	0.4070
Trend GLM (Fixed Year Trend)	Assumes continuing linear macroeconomic drift	0.4028
GLMM: State Random Effects (State RE, addr_state)	Scores future loans with known state geographic adjustments	0.4069
GLMM: Pure Population Marginal (Population Level, re.form = NA)	Scores future loans with zero temporal or regional adjustments	0.4034

Table 11 details out-of-sample performance on the quarantined 2011 temporal holdout. While the GLMM produced a small gain in random-split testing ($0.3815 \rightarrow 0.3897$), **this gain did not persist under strict temporal validation**. The borrower-only GLM achieved a Gini of 0.4070 on 2011 applications, compared with 0.4069 for the state-adjusted GLMM and 0.4034 for the population GLMM. State and temporal effects therefore appear more valuable for portfolio monitoring and risk decomposition than for materially improving future-vintage applicant ranking.

3.4 GLMM Residual Diagnostics via DHARMa Simulation

The retained spatial-temporal GLMM was assessed using 1,000 conditional DHARMa simulations with an explicitly fixed random seed (`simulateREs = "conditional"`). Conditional simulations evaluate residual behaviour within the observed state and origination-month effects. Because the outcome is binary, DHARMa computes randomised quantile residuals, which are expected to follow a continuous standard uniform distribution $\mathcal{U}(0,1)$ under adequate model specification.

However, DHARMa in large-sample binary regressions have limitations; model governance requires recognizing two known statistical limitations when applying simulation-based quantile residuals to large retail portfolios:

1. *Large-Sample Power Paradox ($N > 30,000$):* Standard goodness-of-fit hypothesis tests (such as the Kolmogorov-Smirnov uniformity test and outlier tests) have extreme statistical power in large samples ($N = 30,736$). Consequently, minute, practically trivial departures from theoretical distributions can produce statistically significant p-values ($p < 0.05$), creating a false impression of severe misspecification. Analysts must therefore evaluate effect sizes (such as the dispersion ratio and graphical QQ alignment) rather than relying blindly on p-values.
2. *Bernoulli Discreteness and Overplotting:* For individual binary outcomes ($Y \in \{0,1\}$), raw observation-level quantile-residual plots exhibit severe overplotting and artificial parallel banding, obscuring local curvature and model misspecification. Following DHARMa methodological recommendations (Hartig, 2022), residuals were recalculated across aggregated operational groups, specifically by predicted-risk deciles, geographic jurisdictions, and origination months. These grouped diagnostics assess distinct structural dimensions of model adequacy and complement individual-level tests.

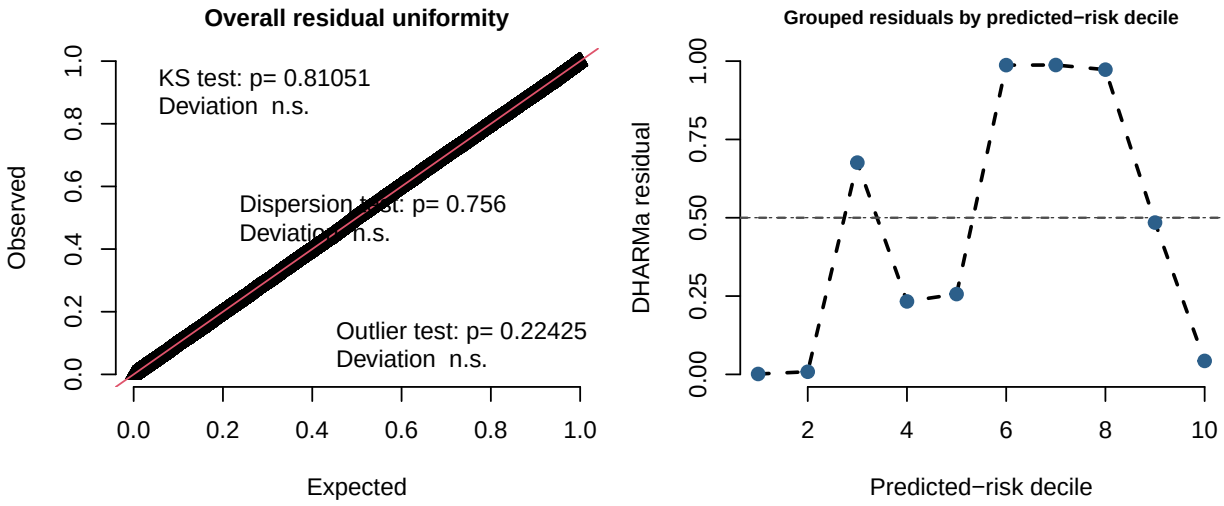


Figure 4: DHARMa diagnostics for the retained spatial-temporal GLMM. The left panel assesses overall scaled-residual uniformity, dispersion, and simulation-envelope outliers using 1,000 conditional simulations. The right panel displays residuals aggregated within ten fitted-risk deciles, with a dashed reference line at the expected median (y = 0.50), an approach recommended for evaluating borrower-level Bernoulli outcomes where individual residual-versus-fitted plots can be visually uninformative. The overall residual diagnostics and grouped risk-decile uniformity test were non-significant. A separate grouped dispersion test identified elevated variability across risk deciles and is discussed in the accompanying text. Temporal dependence between monthly cohorts was assessed separately.

Table 12: DHARMa residual diagnostics

Diagnostic	Result	p-value	Interpretation
Overall residual uniformity	D = 0.0036	0.8105	No overall departure from uniformity detected
Residual dispersion	Ratio = 1.0033	0.7560	No overall overdispersion or underdispersion detected
Simulation-envelope outliers	71 observed; 61.4 expected	0.2243	Non-significant: 0.231% observed vs 0.200% expected
Individual residual quantiles vs fitted probability	Combined p = 0.9405	0.9405	Quantiles approximately horizontal; no conditional distortion
Grouped residual uniformity by risk decile	D = 0.2728	0.3768	No overall uniformity departure detected across 10 risk deciles
Grouped residual dispersion by risk decile	Ratio = 2.5520	0.0200	Elevated variability across risk-decile aggregates

Table 12 summarizes the formal test statistics, p-values, and practical interpretations for the conditional DHARMa residual diagnostics. Figure 4 presents DHARMa scaled-residual diagnostics for the retained spatial-temporal GLMM. The left panel evaluates overall residual uniformity, dispersion, and simulation-envelope outliers using 1,000 conditional simulations. The right panel displays residuals aggregated within ten predicted-risk deciles, with a prominent dashed reference line at the expected median ($y = 0.50$).

Overall borrower-level diagnostics found no significant departure from residual uniformity ($D = 0.0036, p = 0.8105$), dispersion expectations (ratio $= 1.0033, p = 0.7560$), envelope-outlier frequency (71 observed vs 61.4 expected, $p = 0.2243$), or fitted-probability-dependent quantiles (combined $p = 0.9405$). Residuals aggregated across ten predicted-risk deciles also showed no overall uniformity departure ($D = 0.2728, p = 0.3768$), but exhibited greater between-decile variability than expected (ratio $= 2.5520, p = 0.0200$). Appendix C shows that this signal reflects mild overprediction in the lowest risk groups and underprediction in intermediate-to-high risk groups, and that the dispersion result attenuates under finer grouping. State and month aggregates showed no significant uniformity or dispersion departures, although monthly residuals retained serial dependence.

3.4.1 Temporal Autocorrelation on Aggregated Monthly Residuals

Aggregated monthly residuals did not show a significant global uniformity or dispersion departure. However, chronologically ordered monthly residuals exhibited positive serial dependence ($DW = 0.9844, p < 0.001$). This indicates that independent month random intercepts capture month-specific heterogeneity but do not fully represent persistence between adjacent origination cohorts. Possible contributors include macroeconomic persistence, underwriting changes, borrower-composition shifts, outcome seasoning, or other unmeasured time-varying factors. The retained model remains useful for exploratory portfolio decomposition, while a production temporal model could evaluate a continuous smoother or autoregressive covariance structure.

Extended diagnostic investigations, including predictor-specific checks, risk-decile calibration breakdowns, simulation-depth sensitivity across 500, 1,000, and 2,000 runs, and unconditional simulation checks, are documented in Appendix C.

3.5 Variance Partitioning and Intraclass Correlation (ICC)

To evaluate the practical magnitude of jurisdictional and temporal clustering, we extract the variance components from `m4_st_glmm` using `VarCorr()`. In a logistic GLMM, the latent observation-level residual variance is fixed at $\pi^2/3 \approx 3.2899$.

Table 13: GLMM variance components and ICCs

Grouping Level	Variance	Std Dev	Variance Share (ICC)
Jurisdiction (addr_state)	0.0293	0.1711	0.88%
Origination Vintage (issue_d)	0.0183	0.1353	0.55%
Observation-level latent logistic variance	3.2899	1.8138	98.57%
Total Latent Variance	3.3374	1.8269	100.00%

As presented in Table 13, on the latent logistic scale, state and origination-month random intercepts account for estimated variance shares of 0.88% ($\sigma^2_{\text{state}} = 0.0293$) and 0.55% ($\sigma^2_{\text{time}} = 0.0183$), respectively, under the conventional $\pi^2/3 \approx 3.2899$ residual-variance approximation. Most remaining latent variation lies at the observation level rather than between states or months. This calculation does not decompose the variance explained by the measured borrower-level fixed effects.

3.6 Structural Variation: State Caterpillar Plot and Monthly Dynamics

Figure 5 presents a Caterpillar plot displaying Empirical Bayes state random intercepts with **approximate 95% conditional intervals** derived from the conditional variance of the estimated random effects (`condVar = TRUE`), alongside residual monthly dynamics.

3.6.1 Exploratory State-level Findings and Standard Error Reliability

Testing each of the 50 jurisdictions against zero creates a multiplicity consideration; therefore, these intervals represent exploratory state-level deviations rather than definitive regulatory classifications. Exactly **3 states** display approximate 95% conditional intervals that strictly exclude zero:

- **Higher estimated conditional baseline risk:** California ($\hat{b} = +0.231$, 95% CI: $[+0.150, +0.312]$), Florida ($\hat{b} = +0.317$, 95% CI: $[+0.206, +0.428]$), and Nevada ($\hat{b} = +0.433$, 95% CI: $[+0.227, +0.638]$), representing positive estimated random-intercept deviations from the model-wide baseline.

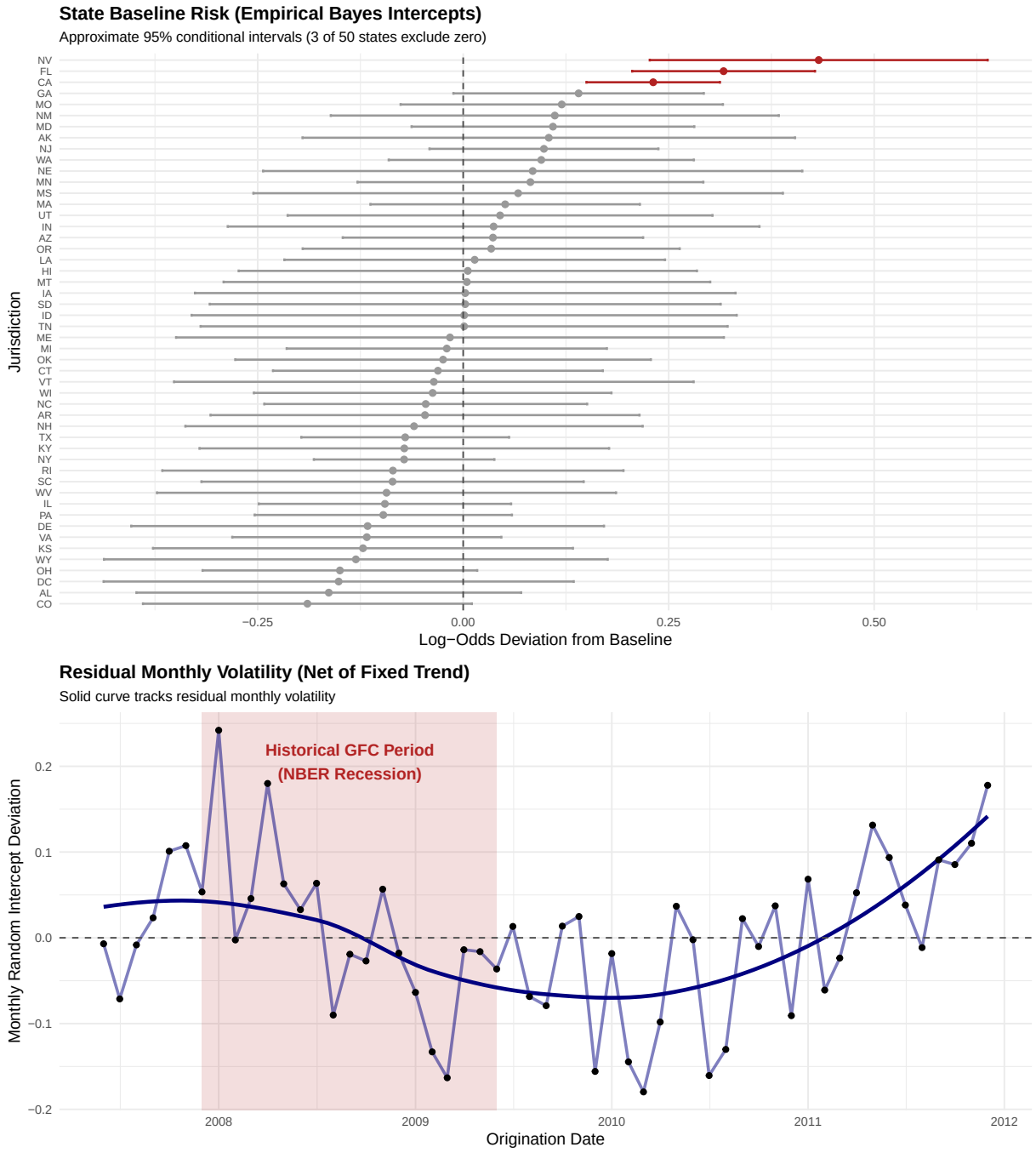


Figure 5: State Empirical Bayes random intercepts (approximate 95% conditional intervals) and residual monthly volatility.

Random effect standard errors and confidence intervals caveat: Model governance requires recognizing that standard errors extracted from `ranef(..., condVar = TRUE)` represent the *posterior conditional variance* of the random effects given the point estimate of the variance component: $\text{Var}(b_i \mid y, \hat{\beta}, \hat{\sigma}^2)$. These frequentist conditional standard errors treat the variance components ($\hat{\sigma}^2_{\text{state}}$ and $\hat{\sigma}^2_{\text{time}}$) as **known, fixed constants without error**. Because the state variance component is small ($\hat{\sigma}^2 = 0.0293$) and estimated across only 50 state clusters, sampling uncertainty in σ^2 is non-trivial. By ignoring hyperparameter uncertainty, the conditional standard errors are artificially optimistic and the resulting Wald-type conditional confidence intervals ($\hat{b}_i \pm 1.96 \times \text{SE}$) are overly narrow (undercoverage). In a fully Bayesian hierarchical specification (e.g., Markov Chain Monte Carlo via `brms`) or a parametric bootstrap where hyperparameter uncertainty is properly integrated, these intervals would widen, potentially causing borderline jurisdictions such as California to encompass zero.

For the remaining 47 jurisdictions, conditional intervals encompass zero. Because 50 jurisdictions were examined, no multiplicity adjustment was applied, and conditional SEs ignore hyperparameter uncertainty, these estimates should be treated as signals for retrospective monitoring rather than definitive geographic risk classifications. **Given the exploratory nature of these state-level estimates, management should refrain from enacting aggressive regional pricing or broad underwriting rejections**, as doing so would penalize applicants based on sampling variation and create severe fair-lending and regulatory compliance exposure.

3.7 Temporal Drift and Macroeconomic Interpretation

The fixed `issue_year` coefficient is $\beta = -0.1508 \pm 0.0384$ ($z = -3.93$, $p < 0.001$). Converted to an Odds Ratio (OR):

$$\text{OR}_{\text{Annual}} = \exp(-0.1508) \approx 0.860 \quad (95\% \text{ CI: } [0.798, 0.927])$$

Holding borrower characteristics and regional clustering constant, each additional origination year is associated with an estimated 14.0% **reduction in default odds**.

In observational lending data, the observed trend may reflect changes in underwriting policy, borrower composition, differential outcome seasoning, macroeconomic conditions, or unmeasured platform practices. The available data cannot identify the relative contribution of these explanations. The model captures empirical drift across historical vintages, but cannot be interpreted as a causal macroeconomic law.

4 Answering Management Questions

4.1 Question 1: Benchmark Comparison and Expected Operational Performance

The newly developed binomial GLM substantially outperformed the supplied legacy benchmark on the common validation sample. Evaluated on the identical validation partition, the new GLM achieved a Gini score of 0.3964 (95% CI: [0.3647, 0.4281]) compared to the legacy benchmark's 0.1100, yielding a substantial net improvement of +0.2864 **Gini points** (+28.6 Gini points on the common validation sample).

On strictly held-out testing data ($N = 7,684$), the model demonstrated generalizable discrimination with a Gini of 0.3735 (AUC = 0.6868, 95% CI: [0.6705, 0.7030]) and a Kolmogorov-Smirnov statistic of 0.280 (measuring the maximum vertical separation between the cumulative score distributions of defaulting and non-defaulting borrowers, achieved at the highest-risk 40.1% of applicants). Under an illustrative 15% probability cutoff scenario, the bank would approve 60.6% of incoming volume while driving the approved portfolio default rate down to 9.2% (compared to the baseline default prevalence of 14.9%).

4.2 Question 2: Variable Importance, Covariate Effects, and Banking Tradition

Dominant Risk Drivers: Backward AIC selection retained nine predictor terms, represented by 32 fitted coefficients including categorical contrasts. The primary drivers of default risk are:

1. *Interest Rate:* Each one-percentage-point increase is associated with approximately 11.7% higher conditional default odds (OR = 1.117, 95% CI: [1.102, 1.132]). This association may partly reflect lender risk-based pricing rather than a purely causal payment-burden effect.
2. *Loan Maturity:* 60-month loans carry 56.4% higher default odds than 36-month loans (OR = 1.564, 95% CI: [1.429, 1.713]).
3. *Loan Purpose:* Small business ventures carry 175.7% higher default odds than auto loans (OR = 2.757, 95% CI: [2.134, 3.562]). Credit card loans (OR = 0.935, $p = 0.589$) show no significant difference from auto loans.
4. *Unreported Employment Tenure:* Applications with unreported employment tenure had approximately 72.9% higher estimated conditional default odds (OR = 1.729, 95% CI: [1.368, 2.185]).
5. *Credit Seeking:* Each recent inquiry increases default odds by 15.3% (OR = 1.153, 95% CI: [1.127, 1.179]).
6. *Public Records:* Each derogatory record increases default odds by 27.9% (OR = 1.279, 95% CI: [1.117, 1.466]).
7. *Revolving Utilization:* Each 10 percentage point increase in credit line utilization raises default odds by 4.3% ($\text{OR}_{10pp} = 1.043$, 95% CI: [1.027, 1.059]).
8. *Annual Income:* Each additional \$10,000 in annual income is associated with approximately 6.6% lower default odds ($\text{OR}_{10k} = 0.934$, 95% CI: [0.923, 0.945]).

Comparison with Banking Sector Tradition (The "5 Cs of Credit"): Traditional retail banking scorecards historically rely on conventional credit metrics that were eliminated during our empirical model selection:

- *Debt-to-Income* (`dti` - Capacity): While banking tradition views DTI as the primary measure of borrower leverage, backward AIC removed `dti` because the model simultaneously incorporates gross income (`annual_inc`), absolute debt (`revol_bal`), and revolving utilization (`revol_util`). The linear signal in `dti` became redundant, with individual liquidity and utilization variables providing superior, disaggregated explanatory power.
- *Historical Delinquency* (`delinq_2yrs` - Character): Traditional scorecards penalize prior 24-month delinquency counts. In this portfolio, active proximate distress signals (recent inquiries `inq_last_6mths`, active derogatory records `pub_rec`, and heavy utilization `revol_util`) captured the predictive signal from historical delinquency.
- *Requested Loan Amount* (`loan_amnt` - Capital/Exposure): Principal size was dropped because default odds are driven by contractual cash-flow burden (captured by `term` and `int_rate`) rather than nominal principal amount in isolation.
- *Credit Depth* (`total_acc`, `open_acc`, `earliest_cr_line`): Account depth offered negligible incremental fit after controlling for active credit strain.
- *Prudential Policy Governance:* While dropped by algorithmic AIC from the statistical PD formula, variables such as DTI, loan size, and past delinquencies must remain codified as hard underwriting policy rules under APRA APS 220 serviceability standards.

4.3 Question 3: Spatial-Temporal Enhancement and Predictive Improvement

Predictive Value of Geography and Time: We explored time through two complementary mechanisms: a **continuous fixed trend** (`issue_year`), capturing long-term structural macroeconomic drift (OR = 0.860, $p < 0.001$), and **discrete monthly random intercepts** (`issue_d`, 55 clusters), capturing residual monthly volatility ($\sigma^2_{\text{time}} = 0.0183$, ICC = 0.55%). Incorporating both state and month random intercepts reduced AIC from 24,001.5 to 23,880.5 and marginally increased random-split conditional test Gini from 0.3815 to 0.3897.

Under strict temporal holdout validation (underwriting 2011 applications with models trained on 2007–2010), the baseline borrower GLM achieves Gini = **0.4070**, while the GLMM with state random effects achieves Gini = **0.4069** and the population GLMM achieves Gini = **0.4034**. State and month random effects improve likelihood-based fit and provide useful portfolio decomposition, but they **do not improve future-vintage**

rank discrimination relative to the borrower-only GLM. Geographic and temporal effects are therefore valuable for portfolio monitoring and risk decomposition, rather than for applicant rank-ordering. Although the retained GLMM is non-singular and passes global distributional and dispersion checks, significant serial dependence remains among aggregated monthly residuals ($DW = 0.9844, p < 0.001$). The model is therefore suitable for exploratory portfolio decomposition but does not fully represent continuous temporal dependence.

4.4 Question 4: Stability of Variable Importance Across Geographic and Temporal Controls

Table 14 examines the stability of the principal borrower-level effects between the baseline GLM and the spatial-temporal GLMM.

Table 14: Stability of principal borrower effects

Predictor	GLM Beta	GLMM Beta	GLM OR	GLMM OR	OR Ratio
term60 months	0.4322	0.5340	1.541	1.706	1.107
int_rate	0.4002	0.3954	1.492	1.485	0.995
emp_lengthUnknown	0.5057	0.5845	1.658	1.794	1.082
annual_inc	-0.4568	-0.4512	0.633	0.637	1.006
inq_last_6mths	0.2218	0.2082	1.248	1.232	0.987
pub_rec	0.0538	0.0533	1.055	1.055	0.999
revol_bal	0.0946	0.0822	1.099	1.086	0.988
revol_util	0.1244	0.1248	1.132	1.133	1.000
purposesmall_business	1.0679	1.0547	2.909	2.871	0.987

Note on Predictor Scales: Continuous and count predictors (`int_rate`, `annual_inc`, `revol_bal`, `revol_util`, `inq_last_6mths`, `pub_rec`) in Table 14 correspond to one-standard-deviation increments in the extended-data analysis and are not directly comparable with the practical-unit odds ratios presented in Table 6, while `term60 months`, `emp_lengthUnknown`, and `purposesmall_business` represent discrete indicator contrasts.

There were **no surprising reversals or dramatic shifts** in variable importance when controlling for regional geography and origination timing. The direction and approximate magnitude of the principal borrower-level effects were structurally preserved between the baseline GLM and the spatial-temporal GLMM. For the primary standardized continuous and count predictors (`int_rate`, `annual_inc`, `revol_util`, `pub_rec`, `revol_bal`, `inq_last_6mths`) as well as the `purposesmall_business` indicator contrast, the odds ratio ratio is between 0.987 and 1.006 (less than 1.5% change). Moderate changes for 60-month term and unknown employment tenure reflect minor confounding with the vintage expansion (60-month loans were introduced in later years) and reporting adjustments. Furthermore, traditional variables dropped by AIC (such as `dti`, `loan_amnt`, and `total_acc`) did not resurrect or become statistically significant when state and time were controlled for. This invariance is an operationally vital result: it demonstrates that applicant-level credit fundamentals operate independently of macroeconomic cycles and state borders, confirming that underwriting scorecards do not require region-specific risk weightings for core borrower attributes.

4.5 Question 5: Structural Risk Variation Across States and Origination Time

Spatial Heterogeneity and Empirical Bayes Shrinkage: Baseline default odds vary across jurisdictions ($\sigma^2_{\text{state}} = 0.0293$), but this variation represents only 0.88% of total latent variance. Exactly 3 states (California, Florida, and Nevada) display approximate 95% conditional intervals excluding zero in this exploratory analysis (with Nevada having the highest estimated deviation $\hat{b} = +0.433$, Florida second $\hat{b} = +0.317$, and California third $\hat{b} = +0.231$). Empirical Bayes shrinkage effectively compressed low-volume states toward the national mean. However, because standard errors extracted from `ranef(..., condVar = TRUE)` treat

variance components as fixed constants without error, these conditional intervals are artificially narrow. Under a full Bayesian framework (e.g., MCMC via `brms`) that accounts for hyperparameter uncertainty, these intervals would widen. Because 50 jurisdictions were examined without multiplicity adjustment and SEs ignore hyperparameter uncertainty, these estimates must be treated as signals for retrospective monitoring rather than definitive classifications. **Given the exploratory nature of these state-level estimates, management should refrain from enacting aggressive regional pricing or broad underwriting rejections.**

Controlling for borrower quality, annual default odds declined by 14.0% annually across the 2007–2011 window (OR = 0.860, $p < 0.001$). The decline may reflect changes in underwriting policy, borrower composition, outcome seasoning, macroeconomic conditions, or other unmeasured platform practices. The analysis cannot identify the contribution of each mechanism.

5 Regulatory Governance, Limitations, and Recommendations

5.1 Compliance and Model Governance

Model Transparency: The logit specification supports model transparency by providing interpretable conditional odds ratios. Production compliance with APRA APS 220 (*Credit Risk Management*) requires independent validation, implementation controls, monitoring, and local-data recalibration.

Post-Selection Inference Mitigation: Held-out validation and testing reduce optimism in reported performance estimates. Conventional coefficient confidence intervals remain conditional on the selected specification and do not encompass model-selection uncertainty, population drift, or implementation risk.

5.2 Limitations

Observational US Dataset: The historical data originate from a US P2P lending platform during 2007–2011. Direct deployment to an Australian retail portfolio requires local redevelopment, independent validation, and recalibration to reflect Australian credit reporting structures, comprehensive credit reporting (CCR) data, superannuation, and local bankruptcy laws.

Absence of Loss Given Default (LGD): The current model predicts Probability of Default (PD) only. True expected credit loss (ECL = PD × LGD × EAD) cannot be finalized until recovery and exposure-at-default models are integrated.

The Future Vintage Dilemma (Scoring Without Unobserved Random Effects): Deploying a GLMM with temporal random intercepts into live underwriting creates an *unobserved vintage problem*: incoming applications in future years (e.g., 2012 or 2026) lack historical random effect estimates ($\hat{u}_t$).

1. *Population Marginal Scoring (re.form = NA):* Setting unobserved shocks to zero ($u_{\text{future}} = 0$) assumes average macroeconomic conditions. Due to Jensen's inequality in non-linear logit models, the expected marginal probability across the random effect distribution is attenuated relative to $\text{logit}^{-1}(X\beta)$:

$$\mathbb{E}_u\left[\text{logit}^{-1}(X\beta + u)\right] \approx \text{logit}^{-1}\left(\frac{X\beta}{\sqrt{1 + c^2\sigma^2}}\right), \quad \text{where } c = \frac{16\sqrt{3}}{15\pi} \approx 0.588$$

   Failing to apply this normal-logistic scaling factor introduces systematic calibration bias.
2. *Autoregressive Forecasting:* Modeling monthly random intercepts as an autoregressive process ($u_t = \phi u_{t-1} + \epsilon_t$) allows short-term forecasting of $\hat{u}_{T+k}$, but introduces out-of-sample macro-forecasting error and procyclicality into underwriting decisions.
3. *Transition to Observable Macroeconomic Covariates:* A superior architectural solution replaces latent random intercepts with explicit, observable macroeconomic indicators (e.g., RBA cash rate, national unemployment rate, CPI inflation, GDP growth), which can be directly evaluated or stressed under APRA APS 220 scenario analysis.

4. *Justification for the Dual Architecture:* This dilemma confirms the operational rationale for our dual-model framework: live loan origination relies on the **borrower-level GLM** (providing deterministic, vintage-invariant, and explainable applicant scoring), while the **spatial-temporal GLMM** is reserved for retrospective portfolio risk decomposition and stress testing within Group Risk.

6 Appendices

6.1 Appendix A: Reproducible R Implementation Script

The principal model specifications are summarised below. The complete executable RMarkdown source accompanies this report for full end-to-end replication:

```
# Core Model Fitting Pipeline
# 1. Benchmark Logistic GLM
formula_sel <- repay_fail ~ term + int_rate + emp_length + annual_inc + purpose +
    inq_last_6mths + pub_rec + revol_bal + revol_util

fit_logit <- glm(formula_sel, data = train, family = binomial(link = "logit"))

# 2. Out-of-sample Evaluation
pred_test <- predict(fit_logit, newdata = test, type = "response")
test_auc  <- pROC::auc(pROC::roc(test$repay_fail, pred_test, quiet = TRUE))
test_gini <- 2 * as.numeric(test_auc) - 1

# 3. Spatial-Temporal GLMM (Random Intercept Model 4)
ext_form <- update(formula_sel, ~ . + issue_year + (1 | addr_state) + (1 | issue_d))
fit_glmm <- lme4::glmer(
  ext_form, data = ext_train, family = binomial(link = "logit"),
  control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e5))
)
```

6.2 Appendix B: Full GLM Parameter Estimates (All Categorical Levels)

Table 15 details the complete parameter estimates for all 32 coefficients in the benchmark GLM, scaled to practical units:

Table 15: Complete GLM parameter estimates

Coefficient	Unit / Scale	Estimate	Std Error	OR	95% CI Lower	95% CI Upper	p-value
Intercept (Baseline)	Baseline constant	-3.5970	0.1387	0.027	0.021	0.036	< 0.001
Loan term: 60 months	vs. 36 months	0.4475	0.0463	1.564	1.429	1.713	< 0.001
Interest rate	Per 1% interest rate	0.1104	0.0068	1.117	1.102	1.132	< 0.001
Employment tenure: 1 year	vs. $<$ 1 year	0.0633	0.0857	1.065	0.901	1.260	0.4605
Employment tenure: 10+ years	vs. $<$ 1 year	0.1148	0.0706	1.122	0.977	1.288	0.1037
Employment tenure: 2 years	vs. $<$ 1 year	-0.1155	0.0822	0.891	0.758	1.047	0.1598
Employment tenure: 3 years	vs. $<$ 1 year	0.0162	0.0827	1.016	0.864	1.195	0.8444
Employment tenure: 4 years	vs. $<$ 1 year	-0.0241	0.0869	0.976	0.823	1.157	0.7813
Employment tenure: 5 years	vs. $<$ 1 year	0.0718	0.0870	1.074	0.906	1.274	0.4092
Employment tenure: 6 years	vs. $<$ 1 year	0.0508	0.0981	1.052	0.868	1.275	0.6046
Employment tenure: 7 years	vs. $<$ 1 year	-0.0797	0.1094	0.923	0.745	1.144	0.4662
Employment tenure: 8 years	vs. $<$ 1 year	0.1562	0.1119	1.169	0.939	1.456	0.1630
Employment tenure: 9 years	vs. $<$ 1 year	-0.0240	0.1226	0.976	0.768	1.241	0.8449

Table 15: Complete GLM parameter estimates *(continued)*

Coefficient	Unit / Scale	Estimate	Std Error	OR	95% CI Lower	95% CI Upper	p-value
Employment tenure: Unknown	vs. < 1 year	0.5474	0.1195	1.729	1.368	2.185	< 0.001
Annual income ($10k)	Per $10,000	-0.0683	0.0060	0.934	0.923	0.945	< 0.001
Purpose: Credit card	vs. car purpose	-0.0675	0.1249	0.935	0.732	1.194	0.5888
Purpose: Debt consolidation	vs. car purpose	0.2029	0.1128	1.225	0.982	1.528	0.0720
Purpose: Educational	vs. car purpose	0.2428	0.2298	1.275	0.813	2.000	0.2905
Purpose: Home improvement	vs. car purpose	0.1584	0.1320	1.172	0.905	1.517	0.2300
Purpose: House	vs. car purpose	0.3228	0.2113	1.381	0.913	2.089	0.1265
Purpose: Major purchase	vs. car purpose	0.0462	0.1451	1.047	0.788	1.392	0.7504
Purpose: Medical	vs. car purpose	0.5944	0.1736	1.812	1.290	2.546	< 0.001
Purpose: Moving	vs. car purpose	0.4892	0.1845	1.631	1.136	2.341	0.0080
Purpose: Other	vs. car purpose	0.4437	0.1226	1.558	1.226	1.982	< 0.001
Purpose: Renewable energy	vs. car purpose	0.3793	0.4098	1.461	0.654	3.262	0.3547
Purpose: Small business	vs. car purpose	1.0141	0.1307	2.757	2.134	3.562	< 0.001
Purpose: Vacation	vs. car purpose	0.4347	0.2254	1.544	0.993	2.402	0.0538
Purpose: Wedding	vs. car purpose	-0.0513	0.1800	0.950	0.668	1.352	0.7754
Credit inquiries (6m)	Per inquiry count	0.1421	0.0114	1.153	1.127	1.179	< 0.001
Derogatory public records	Per record count	0.2463	0.0694	1.279	1.117	1.466	< 0.001
Revolving balance ($10k)	Per $10,000	0.0446	0.0096	1.046	1.026	1.065	< 0.001
Revolving line utilization (10%)	Per 10% points (0.10)	0.0419	0.0081	1.043	1.026	1.059	< 0.001

6.3 Appendix C: Additional GLMM Residual Diagnostics and Sensitivity Checks

6.3.1 C.1. Predictor-Specific Residual Diagnostics

To verify that the continuous functional forms in the spatial-temporal GLMM (Model 4) adequately capture underlying default relationships, Figure 6 plots randomized quantile residuals against the primary continuous financial predictors: standardized interest rate, standardized revolving utilization, logarithmic annual income, and credit inquiry counts.

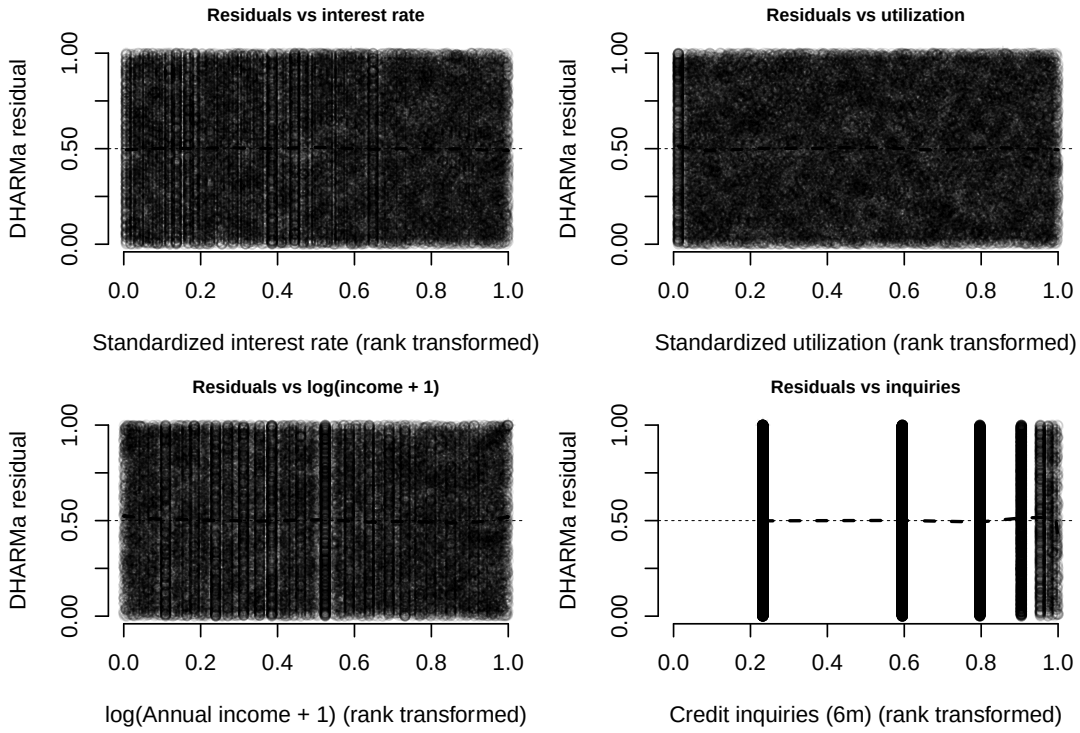


Figure 6: DHARMa randomized quantile residuals plotted against key continuous predictors (semi-transparent scatter with overlaid spline smoothers) assessing functional form.

The fitted spline smoothers remain close to the horizontal line at the nominal median of 0.50 across the operational range of each predictor. This confirms that the linear and logarithmic transformations adequately account for risk variation across interest rates, revolving credit lines, income levels, and inquiry counts without material functional-form distortion.

6.3.2 C.2. Risk-Decile Calibration Breakdown and Grouped Dispersion Analysis

In Section 3.4, the grouped dispersion test across ten predicted risk deciles flagged elevated between-decile variation ($p = 0.0200$). Table 16 profiles conditional fitted-sample calibration and grouped DHARMa scaled residuals across each predicted-risk decile to identify the source of this signal.

Table 16: Conditional fitted-sample calibration and grouped DHARMa residual breakdown

Decile	Applications (N)	Mean Predicted PD	Observed Default Rate	Difference	Binomial SE	Grouped Residual
1	3,074	3.72%	3.16%	-0.57%	0.0034	0.0013
2	3,074	5.71%	5.07%	-0.63%	0.0042	0.0084
3	3,074	7.36%	8.13%	+0.78%	0.0047	0.6760

4	3,074	9.03%	9.37%	+0.34%	0.0052	0.2329
5	3,074	10.83%	11.19%	+0.36%	0.0056	0.2563
6	3,074	12.83%	15.06%	+2.23%	0.0060	0.9872
7	3,073	15.22%	17.64%	+2.42%	0.0065	0.9875
8	3,073	18.41%	20.66%	+2.25%	0.0070	0.9728
9	3,073	23.29%	24.60%	+1.31%	0.0076	0.4846
10	3,073	34.89%	34.88%	-0.00%	0.0086	0.0430

Inspection of Table 16 clarifies the empirical mechanism producing the grouped dispersion signal:

1. **Low-Risk Deciles (1–2):** The model slightly overpredicts default risk (Decile 1: predicted 3.72% vs. observed 3.16%, $\Delta = -0.57\%$; Decile 2: predicted 5.71% vs. observed 5.07%, $\Delta = -0.63\%$). Given the large sample size ($N \approx 3,074$ per decile), these slight negative calibration offsets shift aggregated scaled residuals toward the lower theoretical margin (0.0013 and 0.0084).
2. **Intermediate-to-High Risk Deciles (6–8):** The model modestly underpredicts default risk by 2.2 to 2.4 percentage points (Decile 6: predicted 12.83% vs. observed 15.06%, $\Delta = +2.23\%$; Decile 7: predicted 15.22% vs. observed 17.64%, $\Delta = +2.42\%$; Decile 8: predicted 18.41% vs. observed 20.66%, $\Delta = +2.25\%$). Relative to binomial standard errors (SE ≈ 0.0060 to 0.0070), these deviations correspond to approximately +3.2 to +3.7 standard errors, placing aggregated scaled residuals close to the upper theoretical margin (0.9872, 0.9875, and 0.9728).
3. **Highest-Risk Decile (10):** In the highest-risk decile, predicted default probability (34.89%) closely matches observed default frequency (34.88%, $\Delta = -0.00002$).

Evaluating residual stability across alternative binning resolutions confirms that uniformity is preserved under 5 quintiles ($D = 0.3145, p = 0.6068$), 10 deciles ($D = 0.2728, p = 0.3768$), and 20 ventiles ($D = 0.1957, p = 0.3782$). The dispersion test statistic attenuates as resolution increases, showing significance under 5 groups (ratio $= 3.4593, p = 0.0120$) and 10 groups (ratio $= 2.5520, p = 0.0200$), but becoming non-significant under 20 groups (ratio $= 1.5111, p = 0.1860$). This pattern demonstrates that the signal is driven by coarse group aggregation rather than systematic borrower-level distributional breakdown. Prudential governance requires tracking these intermediate-tier deciles across future out-of-sample vintages.

6.3.3 C.3. Simulation Sensitivity and Outlier Stability

Table 17 evaluates diagnostic sensitivity across three simulation depths ($N_{\text{sim}} \in \{500, 1000, 2000\}$) generated under the conditional simulation configuration.

Table 17: DHARMa diagnostic sensitivity across simulation depths

Simulations	Obs. Outliers	Exp. Outliers	Outlier p	Uniformity p	Disp. Ratio	Dispersion p
500	122	122.7	1.0000	0.6659	1.0027	0.7920
1000	71	61.4	0.2243	0.8105	1.0033	0.7560
2000	27	30.7	0.5874	0.6964	1.0037	0.7080

Across all tested simulation depths in Table 17, the outlier test, uniformity test, and dispersion test show consistent stability with no statistically significant departures from theoretical distributions ($p > 0.05$).

The conditional simulation identified 71 boundary observations compared with 61.4 expected under 1,000 simulations ($p = 0.2243$). Standardized mean differences between outlier accounts and conforming accounts remain small across all key financial indicators ($|\text{SMD}| < 0.25$: annual income SMD $= -0.019$, revolving utilization SMD $= 0.042$, credit inquiries SMD $= 0.081$, interest rate SMD $= 0.088$, and revolving balance SMD $= -0.012$). These small differences indicate no evidence of data corruption or recording errors.

Re-simulating from the fitted random effects distribution (`simulateREs = "user-specified"`, `re.form = NA`, $N = 1,000$) confirmed acceptable uniformity ($D = 0.0065, p = 0.144$) and dispersion (ratio $= 1.0415, p = 0.204$), supporting hierarchical model adequacy.

6.4 Appendix D: Computational Environment and Methodological References

6.4.1 Software Environment and Key Package Versions

Reproducibility was ensured by tracking key package versions utilized in model estimation, diagnostic validation, and performance evaluation (Table 18):

Table 18: Key software packages and analytical environment

Package	Version	Role
lme4	2.0.6	Linear and generalized mixed-effects model estimation via maximum likelihood (bobyqa)
DHARMa	0.5.0	Residual diagnostics and simulation envelopes for generalized linear mixed models
caret	7.0.1	Data partitioning and stratified 5-fold cross-validation
pROC	1.19.1	ROC curve estimation, AUC computation, and DeLong confidence intervals
car	3.1.5	Generalized variance inflation factors (GVIF) for collinearity diagnostics
ggplot2	4.0.3	Data visualization and calibration curve generation
dplyr	1.2.1	Data manipulation, pipeline filtering, and feature transformation
kableExtra	1.4.1	Publication-quality LaTeX table formatting and column specification
knitr	1.52	Dynamic report generation and code execution engine
rmarkdown	2.32	Document conversion pipeline to PDF via pandoc and pdfLaTeX

The underlying analysis was executed in R version 4.6.1 on Linux x86_64 (x86_64-redhat-linux-gnu).

6.4.2 References

1. **APRA Prudential Standard APS 220 (*Credit Risk Management*, effective 1 January 2022; superseding APS 220 *Credit Quality*):** Australian Prudential Regulation Authority (APRA), Sydney. Available at: https://www.apra.gov.au/credit-risk-management. Standards establishing minimum prudential requirements for credit risk governance, independent model validation, and continuous monitoring of credit portfolio performance.
2. **Basel Committee on Banking Supervision (BCBS, 2015):** *Corporate Governance Principles for Banks* and *Guidance on Credit Risk and Accounting for Expected Credit Losses*. Bank for International Settlements (BIS), Basel. Available at: https://www.bis.org/bcbs/publ/d328.htm. International supervisory guidance on credit risk governance, model risk management, and sound credit underwriting.
3. **Hartig, F. (2026):** *DHARMa: Residual Diagnostics for Hierarchical (Multi-Level / Mixed) Regression Models*. R package version 0.5.0. Available at: https://cran.r-project.org/package=DHARMa. DOI: 10.32614/CRAN.package.DHARMa. Comprehensive simulation-based quantile residual framework for generalized linear mixed models.